

FT LEONARD WOOD, MO, USA

WMO: 724457

Lat: 37.740N

Lon: 92.140W

Elev: 1159

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	11.7	-4.2	4.6	8.6	0.0	5.7	15.3	20.6	38.4	18.9	39.2	6.0	320	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	94.8	75.3	91.8	75.3	89.4	75.1	79.1	89.2	77.9	87.7	76.7	86.0	6.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	144.2	84.2	75.1	137.7	83.1	73.9	132.1	81.9	43.5	89.2	42.2	87.9	40.8	85.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.7	16.8	15.1	-1.1	97.5	5.0	3.1	-4.7	99.8	-7.6	101.6	-10.4	103.4	-14.0	105.6		
			-1.8	80.9	4.9	1.5	-5.4	82.0	-8.2	82.9	-11.0	83.8	-14.6	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	32.8	36.7	47.3	57.1	65.3	74.6	77.9	75.3	69.0	57.3	46.2	36.7
	DBStd	18.07	12.30	12.33	11.29	9.21	8.52	5.52	5.42	5.53	7.17	9.17	9.82	10.08
	HDD50	1814	546	400	186	36	4	0	0	0	0	36	181	425
	HDD65	4480	999	793	557	264	106	4	0	2	40	272	565	878
	CDD50	4168	12	28	101	250	480	739	864	783	570	261	68	12
	CDD65	1360	0	1	7	28	117	294	399	320	160	32	2	0
	CDH74	12904	0	6	72	288	1014	2775	4121	2959	1393	263	13	0
	CDH80	5109	0	0	7	56	271	1092	1907	1237	494	45	0	0
	WSAvg	6.9	7.7	7.8	8.4	8.5	6.9	6.0	5.1	5.2	5.5	6.6	7.5	7.4
	PrecAvg	42.7	2.2	2.2	3.6	4.3	5.2	4.3	4.1	3.9	4.3	3.1	3.3	2.6
Precipitation	PrecMax	61.4	5.8	6.0	10.0	11.9	11.4	7.7	10.4	14.1	13.5	10.8	9.1	10.7
	PrecMin	25.5	0.5	0.2	1.2	0.6	1.0	0.4	0.7	0.8	0.4	0.2	0.4	0.6
	PrecStd	9.8	1.3	1.6	1.9	2.2	2.4	1.9	2.3	2.4	2.8	2.0	2.4	2.1
	PrecStd	9.8	1.3	1.6	1.9	2.2	2.4	1.9	2.3	2.4	2.8	2.0	2.4	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	73.6	80.8	86.1	89.7	95.2	100.4	97.4	92.7	84.9	76.1	67.2
		MCWB	56.8	59.8	62.6	66.3	70.1	74.7	73.5	75.5	73.7	69.0	61.5	60.2
	2%	DB	62.4	66.9	75.2	80.9	85.8	91.2	96.2	93.5	89.4	80.3	70.7	62.3
		MCWB	53.4	56.3	60.2	63.7	70.6	76.4	74.8	75.2	74.1	65.3	57.7	56.4
	5%	DB	56.4	62.1	69.9	77.2	83.1	89.0	92.5	90.6	86.2	76.9	66.3	57.1
		MCWB	48.4	53.3	57.3	61.7	69.5	75.6	76.1	74.2	73.0	64.1	56.6	49.7
	10%	DB	51.1	56.3	65.6	73.3	80.1	86.6	89.6	87.2	82.3	72.6	62.4	52.8
		MCWB	44.1	49.0	54.5	60.6	68.1	74.4	76.1	73.9	70.6	62.0	53.7	47.0
	0.4%	WB	61.1	62.3	65.8	69.7	75.0	80.2	80.7	80.5	77.3	72.3	64.9	62.4
		MCDB	65.5	70.6	75.7	79.7	84.0	88.9	89.5	91.2	86.8	80.4	70.9	66.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	55.3	59.4	62.7	67.3	73.2	78.3	79.2	78.6	75.5	69.4	61.5	57.9
		MCDB	58.9	65.8	71.7	75.0	82.3	87.3	89.3	89.2	85.2	75.6	66.5	61.0
	5%	WB	49.8	55.5	59.8	65.3	71.4	77.1	78.0	77.0	74.2	66.7	58.6	52.4
		MCDB	54.5	59.5	67.8	73.0	79.1	85.9	88.1	86.8	83.2	72.7	64.2	55.6
	10%	WB	44.5	49.3	56.8	62.9	70.1	75.5	76.9	75.3	72.7	64.4	55.0	47.8
		MCDB	51.1	55.5	63.1	70.1	77.5	84.0	86.4	84.0	80.6	70.9	61.2	51.5
Mean Daily Temperature Range	5% DB	MDBR	19.1	19.1	20.7	21.7	20.3	19.3	19.9	19.7	20.5	21.5	20.1	17.3
		MCDBR	25.3	26.4	26.8	26.0	23.0	21.7	23.6	24.0	22.8	24.0	25.0	24.3
		MCWBR	17.1	17.7	13.8	13.0	10.2	9.5	7.6	8.4	9.4	11.9	15.2	16.7
	5% WB	MCDBR	22.9	23.2	23.2	21.7	19.4	19.4	19.6	20.0	20.0	20.0	21.2	21.4
		MCWBR	18.6	17.9	13.8	13.3	9.8	9.8	8.3	8.8	9.1	11.9	16.1	18.2
Clear-Sky Solar Irradiance	taub		0.300	0.308	0.336	0.384	0.415	0.426	0.452	0.450	0.386	0.344	0.309	0.295
	taud		2.482	2.457	2.426	2.298	2.272	2.264	2.223	2.222	2.378	2.486	2.532	2.517
	Ebn at Noon		282	294	294	284	276	272	263	261	274	277	277	277
	Edh at Noon		25	30	34	41	43	43	44	43	35	29	24	23
	RadAvg		719	921	1231	1595	1793	2045	1989	1779	1534	1118	778	608
All-Sky Solar Radiation	RadStd		74	107	104	97	157	157	135	100	104	122	82	69

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page